SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **K Ganesh Kumar/192365016** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

K Ganesh Kumar
Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
   a. Show differences and similarities
      i. Map how each affects the Data structure and Security level
      ii. Analyze which is more vulnerable to attacks and why Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
   b. Show relationships between these concepts
   c. Analyze how each contributes to secure communication
   d. Include one real-world application for each
4. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
   a. Show how each concept is mathematically connected
   b. Map their role in encryption/decryption
   c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understandi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technolo gies	Good understandi ng with proper integration of most concepts	Basic understandin g; limited or partial integration	Poor understandin g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementat ion with minor issues	Basic implementati on with errors or inefficiencies	Incorrect or incomplete implementati on
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretatio n	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentat ion	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentati on with necessary details	Basic documentatio n with missing details	Poor or incomplete documentatio n

Question 1: Security Attacks, Threats, Risks and Security Goals
Extended Concept Map

INFORMATION SECURITY

Security Attacks Security Threats Security Risks

▼ ▼ ▼ ▼ ▼ ▼ ▼ ▼ ▼ Financial Reputation
Passive Active Malware Unauthorized Data Data Legal & Loss Damage
Attack Attack Attack Access Theft Leakage Compliance

▼ ▼ ▼
Sniffing DoS Attack Virus
Monitoring MITM Attack Worm
Traffic Modification Trojan
Analysis Ransomware Spyware

▼
Consequences

▼ ▼ ▼ Data Loss Business Loss
Privacy Breach
▼ SECURITY
GOALS

▼ ▼ ▼ Confidentiality
Integrity Availability
▼ ▼
Encryption Hashing ▼
Backup
Access Ctrl Digital Sign Disaster Recovery
Authentication Checksums Load Balancing
▼ SECURE INFORMATION SYSTEM

Features Security Attacks

- Attempt to compromise information systems.
- Can be Passive or Active.
- Target Confidentiality, Integrity or Availability.

Security Threats

- Unauthorized Access
- Data Theft
- Malware Infection
- Insider Threat
- Social Engineering

Security Risks

- Financial Loss
- Reputation Damage
- Legal Issues
- Customer Trust Loss

- Operational Downtime

Security Goals

Confidentiality

- Prevent unauthorized access
- Encryption
- Authentication

Integrity

- Prevent data modification
- Hash Functions
- Digital Signatures

Availability

- Ensure continuous access
- Backup
- Redundancy
- Disaster Recovery

Real World Examples

Attack	Threat	Risk	Security Goal
Phishing	Password Theft	Bank Fraud	Confidentiality
SQL Injection	Database Modification	Wrong Reports	Integrity
DDoS	Server Crash	Website Down	Availability
Ransomware	Data Encryption	Business Shutdown	Availability
Insider Attack	Information Leak	Reputation Damage	Confidentiality

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Analysis

Passive attacks mainly affect **Confidentiality**, whereas active attacks compromise **Integrity** and **Availability**. Organizations implement encryption, authentication, hashing, firewalls, IDS/IPS, backups, and disaster recovery plans to reduce these risks. Together, the CIA triad forms the foundation of modern cybersecurity.

Question 2: Concept Map on Substitution Techniques

Question

Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Connect them based on:

- **Key Usage**
- **Complexity**
- **Security Strength**

Analyze which technique is the most secure and justify your answer.

Introduction

Substitution ciphers are one of the earliest encryption techniques in cryptography. In these methods, each letter of the plaintext is replaced with another letter or symbol according to a predefined rule or key. Although these techniques are considered classical encryption methods, they form the basis for understanding modern cryptographic algorithms.

The four major substitution techniques are:

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher
- Hill Cipher

Each technique differs in terms of key usage, encryption complexity, and security strength.

Extended Concept Map

SUBSTITUTION TECHNIQUES

Caesar Cipher Monoalphabetic Cipher Polyalphabetic Cipher

▼ Vigenère Cipher

Fixed Shift Key Random Alphabet Mapping Repeating Keyword

Easy Encryption Moderate Encryption Multiple Alphabets

▼ Hill
Cipher
Matrix Key (Linear
Algebra)

▼

Multiple Letter Encryption

▼

Highest Security Among Classical Ciphers

Relationship Between the Techniques

SUBSTITUTION CIPHERS

Caesar Monoalphabetic Vigenère Hill

Single Shift Random Mapping Repeating Key Matrix Key

Low Complexity Medium Complexity High Complexity Very High Complexity

Weak Security Moderate Security Strong Security Strongest Security

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Key Usage Comparison

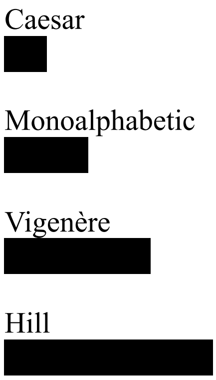
Cipher	**Key Used**	**Description**
Caesar Cipher	Integer Shift	Every letter is shifted by a fixed number of positions.
Monoalphabetic Cipher	Random Alphabet	Each letter is permanently mapped to another alphabet.
Vigenère Cipher	Keyword	The keyword repeats until it matches the plaintext length.
Hill Cipher	Matrix Key	Uses matrix multiplication for encryption and decryption.

Complexity Comparison

Cipher	**Encryption Method**	**Complexity**
Caesar	Alphabet Shift	Low
Monoalphabetic	Letter Mapping	Medium
Vigenère	Multiple Alphabets	High
Hill	Matrix Operations	Very High

Security Strength Comparison

Security Strength



Cipher	Security Level	Reason
Caesar	Weak	Only 25 possible keys.
Monoalphabetic	Moderate	Larger key space but vulnerable to frequency analysis.
Vigenère	Strong	Uses multiple substitution alphabets.
Hill	Very Strong	Encrypts blocks of letters using matrix mathematics.

Advantages of Each Cipher

Caesar Cipher

- Very easy to understand.
- Simple implementation.
- Fast encryption and decryption.
- Good for learning basic cryptography.

Monoalphabetic Cipher

- Larger key space than Caesar.
- More difficult to guess.
- Better resistance against brute-force attacks.

Vigenère Cipher

- Uses multiple alphabets.
- Prevents simple frequency analysis.
- More secure than Caesar and Monoalphabetic.

Hill Cipher

- Uses matrix multiplication.
- Encrypts multiple characters together.
- Strong resistance against frequency analysis.
- Suitable for block encryption.

Limitations

Cipher	Limitation
Caesar	Easily broken using brute force.
Monoalphabetic	Frequency analysis can reveal plaintext.
Vigenère	Vulnerable if the keyword is short or repeated.
Hill	Requires matrix calculations and invertible keys.